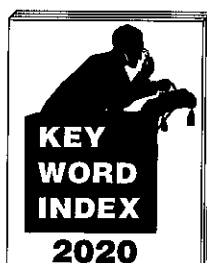


2023

JOURNEYMAN
OPEN
BOOK
EXAM #3



25 QUESTIONS
TIME LIMIT - 1 HOUR

TIME SPENT **MINUTES**

SCORE **%**

JOURNEYMAN OPEN BOOK EXAM #3 25 Questions One Hour Time Limit

1. As a general rule, receptacle outlets provided for the small appliance countertop circuits in the kitchen of a dwelling, on or above countertop or work surfaces: On or above, but not more than ____ above, the countertop or work surface.
- (a) 12" (b) 18" (c) 20" (d) 24"
2. For a feeder tap not exceeding 10 feet in length and field installation, the maximum overcurrent device rating supplying a tap conductor with an ampacity of 40 amps is ____.
- (a) 150a (b) 200a (c) 350a (d) 400a
3. In dwelling units, hallways of ____ feet or more in length shall have at least one receptacle outlet.
- (a) 6 (b) 8 (c) 10 (d) 12
4. Where a 20 amp receptacle is installed in a residential laundry area, designated for the washing machine, the receptacle shall be provided with ____ protection.
- (a) GFCI (b) AFCI (c) both GFCI and AFCI (d) neither GFCI and AFCI
5. In guest rooms of a hotel or motel, the receptacle outlets shall be permitted to be located conveniently for permanent furniture layout. At least ____ receptacle outlet(s) shall be readily accessible.
- (a) one (b) two (c) three (d) four
6. For a dwelling unit service calculation, a demand factor of 75% may be applied to the nameplate of four or more appliances rated 1/4 hp or ____ watt or greater if fastened in place.
- (a) 500 (b) 650 (c) 750 (d) 1,000
7. Thermostatically controlled switching devices serving both controllers and disconnecting means for fixed electric space heating equipment shall ____.
- (a) be not permitted
(b) be located not more than 8 feet above floor level
(c) open all grounded conductors when placed in the off position
(d) be designed so that the circuit cannot be energized automatically after the device has been manually placed in the off position

8. Where conduits enter a floor-standing switchboard, switchgear or, panelboard at the bottom, the conduits, including their end fittings, shall not rise more than _____ inches above the bottom of the enclosure.

(a) 2 (b) 3 (c) 4 (d) 6

9. Fuel dispensing systems shall be provided with one or more clearly identified emergency shutoff devices or electrical disconnects. Such disconnects or devices shall be installed in approved locations but not less than 20 feet or more than _____ feet from the fuel dispensing devices that they serve.

(a) 50 (b) 75 (c) 80 (d) 100

10. The ampacity of three #10 THWN-2 conductors installed in a raceway is _____ amps if the ambient temperature is 112°F.

(a) 31.6 (b) 34.8 (c) 35 (d) 37.2

11. The conductor is marked RHW-2 on the insulation, what does the -2 represent?

- (a) The cable has two conductors.
- (b) The conductor is double insulated.
- (c) The conductor is thermoplastic.
- (d) The conductor has a maximum operating temperature of 90°C.

12. Outlets supplying all permanently installed nonsubmersible pump motors rated 250 volts or less and _____ amperes or less, single- or 3-phase, shall be provided with GFCI protection.

(a) 60 (b) 50 (c) 30 (d) 20

13. For an arc welder with an I_{max} rating of 43 amps, what is the maximum rating for the overcurrent device supplying the welder?

(a) 85 amp (b) 90 amp (c) 95 amp (d) 100 amp

14. In dwelling units, all _____ volt receptacles in locations specified and supplied in single-phase branch circuits rated 150 volts or less to ground shall have GFCI protection for personnel.

(a) 110-220 (b) 115-230 (c) 120-240 (d) 125-250

15. All 15 or 20 amp, single-phase, 125 volt through 250 volt receptacles located within _____ feet of a fountain edge shall be provided with GFCI protection.

(a) 20 (b) 24 (c) 25 (d) 30

16. Where ungrounded conductors are run in parallel in multiple raceways, the equipment grounding conductor, where used, shall be ____.

- (a) installed in one raceway only (b) run in parallel in each raceway
(c) not required (d) bare

17. Where a dwelling has a two car attached garage, the Code requires a minimum of ____ receptacle outlet(s) shall be installed in the garage.

- (a) none required (b) one (c) two (d) three

18. Track lighting where installed in a continuous row, each individual section of not more than ____ feet in length shall have one additional support.

- (a) 4 (b) 6 (c) 8 (d) 12

19. The demand load for a 14 kW range installed in a dwelling unit is ____ kW.

- (a) 8 (b) 8.4 (c) 8.8 (d) 12

20. Where multiple driven ground rods are used to form the grounding electrode system, in order to maintain an effective grounding electrode system, they shall not be less than ____ apart.

- (a) 36" (b) 48" (c) 60" (d) 72"

21. Luminaires installed in exposed or concealed locations under metal-corrugated sheet roof decking shall be installed and supported so there is not less than ____ clearance from the lowest surface of the roof decking to the top of the luminaire.

- (a) 1 1/4" (b) 1 1/2" (c) 1 3/4" (d) 1 7/8"

22. Duty on elevator and dumbwaiter driving machine motors and driving motors of motor-generators used with generator field control shall be rated as ____.

- (a) intermittent (b) lockable (c) continuous (d) varying

23. The frame of a portable generator at a tree lot shall ____ to be connected to a grounded electrode system if the normally non-current carrying metal parts of equipment and the EGC terminals of receptacles are bonded to the generator frame, and the generator only supplies receptacles mounted on the generator.

- (a) not be required (b) be required (c) not be permitted (d) be permitted

24. Where abandoned communications cables are identified for future use with a tag, the tag shall be ____.

(a) identified (b) suitable (c) marked (d) of sufficient durability to withstand environment

25. When an installation uses metal conduits entering service equipment or enclosures with concentric or eccentric knockouts, the raceway(s) shall be equipped with ____.

(a) single locknuts

(b) double locknuts

(c) at least one locknut and one bushing

(d) bonding jumpers, and bonding-type locknuts or bushings